CLAMAV UPX UNPACKER — CLAM_UPX_TEST

UPX Version Capability Reference

Verified against UPX source: 1.20 · 2.02 · 3.09 · 4.2.0 · 5.0.1 · 5.1.1

Default Method Selection Logic

BY VERSION RANGE

BASE METHOD	FETCH WIDTH	USED BY
getDefaultCompressionMethods_le32()	LE32	win32/pe, win64/pe, linux/i386 ELF, linux/amd64 ELF, djgpp2, watcom/le, most vmlinux
getDefaultCompressionMethods_8()	8-bit	linux/arm ELF, linux/arm64 ELF, linux/mips ELF, linux/riscv64 ELF, macho/ARM, ps1/exe (32-bit)

METHOD → CLI FLAG MAPPING

FORMAT CLASS	CUSTOM METHOD LIST	REASON
PackDos16Com	`{M_NRV2B_LE16, M_END}`	DOS COM; 16-bit fetch only, NRV2B only
PackMachDylib (amd64)	`{M_NRV2E_LE32, M_END}`	dylib; single method forced
PackWinCeArm	`{M_NRV2B_8, M_NRV2E_8, M_LZMA, M_END}`	WinCE ARM; explicit 3-method list

CLI Flag Reference

FLAG AVAILABILITY BY VERSION

FLAG	OPTION ID	FIRST AVAILABLE	EFFECT
--nrv2b	702	2.02	Forces `opt->method = M_NRV2B_LE32` (or _8 depending on format)

FLAG	OPTION ID	FIRST AVAILABLE	EFFECT
--nrv2d	704	2.02	Forces `M_NRV2D`
--nrv2e	705	2.02	Forces `M_NRV2E`
--lzma	721	4.20	Forces `M_LZMA`
--no-lzma	722	4.20	Disables LZMA from all-methods expansion
--prefer-ucl	724	3.09+	Sets `opt->prefer_ucl = true` ; routes NRV calls to UCL library (identical output)
--prefer-nrv	723	3.09+	Clears `prefer_ucl` (default state)
--all-methods	524	2.02	Sets `opt->all_methods` ; expands method list to full set for the format. Removed from help in 3.09+, superseded by --brute.
--brute	—	3.09	Tries all methods × all filters. Equivalent to `M_ALL` expansion excluding M_ULTRA_BRUTE entries.
--ultra-brute	—	3.09	Adds M_LZMA_003 and M_LZMA_407 LZMA variants to the search space.
-1 … -9	—	1.20	Compression level. Level 1 forces NRV2B regardless of file size.

Version Capability Matrix

RUNTIME CAPABILITY BY VERSION

CAPABILITY	1.20	2.02	3.09	4.20	5.01	5.11
NRV2B compression	✓	✓	✓	✓	✓	✓
NRV2D compression	✓	✓	✓	✓	✓	✓
NRV2E compression	—	✓	✓	✓	✓	✓
LZMA compression	—	—	✓	✓	✓	✓
Method CLI flags	—	✓	✓	✓	✓	✓
--lzma flag	—	—	—	✓	✓	✓
win64/pe (64-bit PE)	—	—	—	✓	✓	✓

CAPABILITY	1.20	2.02	3.09	4.20	5.01	5.11
linux/amd64 ELF	—	✓	✓	✓	✓	✓
linux/arm64 ELF	—	—	—	✓	✓	✓
linux/riscv64 ELF	—	—	—	—	—	✓
--brute / --ultra-brute	—	—	✓	✓	✓	✓
zstd (staged, inactive)	—	—	—	staged	staged	staged
bzip2 (staged, inactive)	—	—	—	—	staged	staged

Version Selection Guidance

TARGET → MINIMUM VERSION

REQUIRED METHOD	MINIMUM VERSION	NOTES
--nrv2b	2.02	Present in 1.20 but not flag-selectable. Use 2.02+ if explicit flag required.
--nrv2d	2.02	Same as NRV2B.
--nrv2e	2.02	Stub absent in 1.20 — will fail silently or error.
--lzma	4.20	Stubs present from 3.09 but --lzma CLI flag not exposed until 4.20.
win64/pe target	4.20	No win64 packer class before 4.0.
linux/arm64 ELF target	4.20	arm64 stubs absent in 3.09.
linux/riscv64 ELF target	5.10	Source-confirmed in 5.1.1; first appeared 5.1.0 per release notes.

RECOMMENDED TEST VERSIONS

VERSION	ANOMALY
1.20	No method flags; NRV2E absent; win64/pe and linux/amd64 absent. Will silently fall back to NRV2B or NRV2D based on size heuristic.
2.02	LZMA absent. --lzma flag will error. linux/amd64 present but NRV2E stub may be limited to certain arch combinations — confirm against specific target before relying on

VERSION	ANOMALY
	it.
3.09	LZMA stubs present for all arches but no --lzma CLI flag. LZMA can only be invoked via --brute/--ultra-brute or by patching opt->method directly. win64/pe absent.
4.20 / 5.01 / 5.11	Full capability. Differences between these three are minor (riscv64, staged backends, bug fixes). For a stable test corpus, 4.20 is the oldest version covering all major use cases.

Version Comparison

PE FORMAT SUPPORT

VERSION	RELEASE DATE	GAP	SOURCE INGESTED	NOTABLE ADDITION
1.20	May 23, 2001	—	yes	NRV2B + NRV2D only; win32/pe; linux/386; no NRV2E; no CLI method flags
2.02	Aug 13, 2006	+5 yr	yes	NRV2E stubs; linux/amd64 64-bit ELF; --nrv2b/d/e flags; --all-methods
3.09	Feb 18, 2013	+7 yr	yes	LZMA stubs across all arches; UCL/zlib backend infrastructure; --brute/--ultra-brute
4.20	Oct 26, 2023	+10 yr	yes	win64/pe; --lzma CLI flag; arm64 ELF stubs; zstd backend staged
5.01	May 6, 2025	+2 yr	yes	bzip2 backend staged; lzma-multi stub (i386); --fileinfo
5.02	Jul 20, 2025	+2 mo	—	Bug fixes only
5.10	Jan 7, 2026	+6 mo	—	linux/riscv64 ELF added (release notes only — not ingested)
5.11latest	Mar 5, 2026	+2 mo	yes	riscv64 NRV2B/2D/2E + LZMA stubs; MIPS shared-object stubs; auipc filter (RISC-V); arm.v5a ELF stub

ELF FORMAT SUPPORT

FORMAT	1.202001	2.022006	3.092013	4.202023	5.012025	5.112026
Windows PE						

FORMAT	1.202001	2.022006	3.092013	4.202023	5.012025	5.112026
win32/pe (32-bit)	✓	✓	✓	✓	✓	✓
win64/pe (64-bit)	—	—	—	✓	✓	✓
Linux ELF						
linux/i386 (32-bit)	✓	✓	✓	✓	✓	✓
linux/amd64 (64-bit)	—	✓	✓	✓	✓	✓
linux/arm64 (64-bit)	—	—	—	✓	✓	✓
linux/riscv645.10	—	—	—	—	—	✓

COMPRESSION METHOD SUPPORT

METHOD / ROLE	1.202001	2.022006	3.092013	4.202023	5.012025	5.112026
NRV family — primary compression; decompressor stub embedded in packed binary						
NRV2Bl_lx_n2b.h → nrv2b_d*.S	✓	✓	✓	✓	✓	✓
NRV2Dl_lx_n2d.h → nrv2d_d*.S	✓	✓	✓	✓	✓	✓
NRV2Enrv2e_d*.S	—	✓	✓	✓	✓	✓
LZMA — primary compression; decompressor stub embedded in packed binary						
LZMAlzma_d*.S / compress_lzma.cpp	—	—	✓	✓	✓	✓
UCL — NRV fallback; identical algorithm and wire format, different library						
UCLcompress_ucl.cpp / vendor/ucl/	—	—	fallback	fallback	fallback	fallback
zlib — utility only; provides adler32/crc32, never invoked as compressor						
zlibcompress_zlib.cpp	—	—	checksum only	checksum only	checksum only	checksum only
zstd / bzip2 — staged; full implementation present, M_ constants marked NOT YET USED in conf.h						
zstdcompress_zstd.cpp / M_ZSTD=16	—	—	—	staged	staged	staged
bzip2compress_bzip2.cpp / M_BZIP2=17	—	—	—	—	staged	staged

METHOD / ROLE	1.202001	2.022006	3.092013	4.202023	5.012025	5.112026
CLI flags						
--nrv2b / --nrv2d / --nrv2e	—	✓	✓	✓	✓	✓
--lzma	—	—	—	✓	✓	✓
--all-methods	—	✓	—	—	—	—
--brute / --ultra-brute	—	—	✓	✓	✓	✓

Derived from source analysis of six UPX release trees (1.20–5.1.1). Method constants frozen since 3.09. See clam_upx_test for full test corpus.